The distributive law

1 A student incorrectly used the distributive property and wrote $6\left(7x+2\right)=42x+2$.

Explain how to correct the error.

2 Expand the following expressions:

a $5\left(s+7\right)$	**b** $9\left(5+w\right)$	**c** $3\left(r-6\right)$	**d** $\left(y+8\right)\times 7$
e $-8\left(t+10\right)$	**f** $-8\left(c-5\right)$	**g** $-\left(y+2\right)$	**h** $4\left(4s+5\right)$
i $-10\left(8+3r\right)$	**j** $-8\left(-2t-3\right)$	**k** $p\left(q+4\right)$	**l** $9\left(u-v\right)$
m $-\left(2m-3n\right)$	**n** $-r\left(11s+13t\right)$	**o** $4\left(2r+3s+5t\right)$	**p** $6\left(4m+3n+5p\right)$

3 Expand the following expressions:

a $6m\left(2n+3\right)$	**b** $9\left(v^2+6\right)$	**c** $r\left(r+7\right)$	**d** $6m\left(4m+7\right)$
e $-6a\left(2a+9\right)$	**f** $4wy\left(y+w\right)$	**g** $4p\left(4q-3+6r\right)$	**h** $2f\left(f-7g-6h\right)$
i $-10\left(y^2+7\right)$	**j** $-\left(9-s^3\right)$	**k** $r\left(s^2+8\right)$	**l** $-10\left(3+8u^3\right)$
m $4s^5\left(4t^4-9\right)$	**n** $6p^3\left(2q^5-9\right)$	**o** $9f\left(f-4g-6h\right)$	**p** $3uv\left(v+u\right)$

4 Expand and simplify the following expressions:

a $4\left(x+8\right)-2$	**b** $7+5\left(3x+4\right)$
c $5+3\left(x+4\right)$	**d** $8\left(3x+4\right)-6x$
e $34+9\left(4x-5\right)$	**f** $5x-8\left(2x-3\right)$
g $9x-5\left(2x+3\right)$	**h** $-8\left(5x-7\right)-9$
i $4y+5+6\left(y-9\right)$	**j** $4\left(5\left(x-3\right)+2\right)$

5 Expand and simplify the following expressions:

a $x\left(x+4\right)+8\left(x+9\right)$	**b** $8y^2-4y+7y\left(y-5\right)$
c $3\left(6x-5y\right)-2\left(9y+4x\right)$	**d** $7\left(y+9\right)+5\left(y+6\right)$
e $10\left(x-4\right)+3\left(x+8\right)$	**f** $5\left(y+7\right)-\left(y-9\right)$
g $-8\left(x+4\right)-3\left(x-2\right)$	**h** $11\left(a+4\right)-6\left(4-a\right)$
i $4u\left(6u+9v\right)-2u\left(8v+5u\right)$	**j** $9w\left(8w+6z\right)-5w\left(4z+2w\right)$

Applications

6 A solar energy company offers a range of solar panels, each with a power rating of

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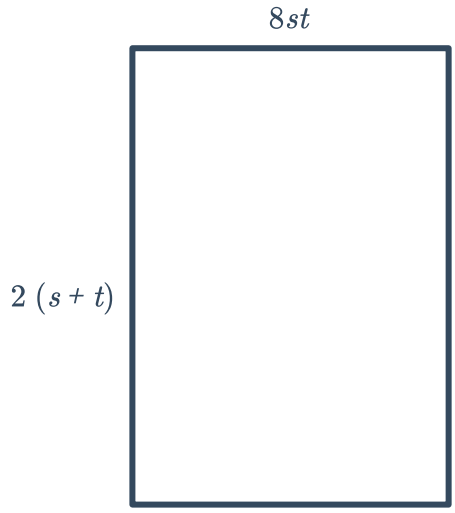
$(m + 8)$ kW, where m is the panel model number. Write an expression in terms of m for the total amount of power produced by 11 solar panels.

7 The representatives of 8 countries are attending a global summit, and each country is able to send $(2r - 3)$ representatives. Write an expression for the total number of representatives attending the summit.

8 A forklift operator has enough time to make 5 trips between the two ends of a warehouse. A forklift can carry $(2 + 3n^4)$ boxes at a time. Write an expression for the total number of boxes that can be carried across the warehouse.

9 Consider the rectangle shown:

Find an expression for the area of the rectangle in expanded form.



10 A rectangle is 8 m longer than it is width. Its width is represented by the pronumeral u.

Form an expression for the area of the rectangle in expanded form.

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